

**MIRPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY AZAD
KASHMIR**



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Lab Task: Software Quality Engineering #3

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EEE 829 TEST PLAN: UNIVERSITY LMS

Document ID: SQE_Lab3_TahirKhan_TestPlan

Project: Library Management System (LMS)

Date: April 29, 2026

1. Test Plan Identifier

LMS-TP-2026-001. This plan supports the first major release of the University Library Management System, specifically focusing on core student-facing modules.

2. Introduction

The LMS is a centralized platform for managing university library resources. This report outlines the verification process for **Authentication (M1)**, **Search (M2)**, **Loans (M3)**, and **Notifications (M4)**. It ensures that the software transitions from the **Test Case Design** phase to **Execution** with a clear roadmap for quality assurance.

3. Test Items

The following artifacts and modules are subject to testing:

- **Web Application:** Front-end built with HTML/CSS and JavaScript.
- **Backend Server:** Node.js/Express API.
- **Database:** MySQL tables (Users, Books, Loans, Fines).
- **External APIs:** SMTP Server (for emails) and SMS Gateway.

4. Features to be Tested

Each feature is mapped to a specific requirement for traceability:

- **Security (REQ-01 to REQ-04):** Verification of the hashing algorithm for passwords and the "lockout" logic (3 failed attempts triggering a 15-minute freeze).
- **Inventory Search (REQ-05 to REQ-08):** Accuracy of the MySQL LIKE queries for Title/Author and verification of pagination logic for datasets exceeding 20 records.
- **Business Logic (REQ-09 to REQ-12):** Mathematical verification of the 14-day loan duration and the incrementing fine logic (\$1.00 per day overdue).

- **Alert System (REQ-13 to REQ-16):** Triggering of Cron jobs for 48-hour pre-due-date email alerts.

5. Features Not to be Tested

- **Mobile Responsiveness:** Only Desktop views (Chrome/Firefox) are in scope for Lab 3.
- **Admin Dashboard:** Backend inventory management (adding/deleting books) will be tested in Phase 2.
- **Network Latency:** Performance is tested on a local/staging environment, not across high-latency satellite connections.

6. Approach (Testing Strategy)

We will utilize the **Software Testing Life Cycle (STLC)** framework:

- **Unit Testing:** Verified by developers via Jest.
- **System Testing (Our Focus):** End-to-end testing of user flows.
- **Equivalence Partitioning (EP):** Grouping valid usernames (e.g., FA23-BSE-001) and invalid ones to reduce redundant test cases.
- **Boundary Value Analysis (BVA):** Testing the system at exactly 14 days, 15 days, and 0 days for loan calculations.

7. Item Pass/Fail Criteria

- **Pass:** The actual database state matches the expected state (e.g., `is_issued` changes from 0 to 1 upon loan).
- **Fail:** UI elements not loading within 2 seconds (REQ-07) or logic errors in fine calculation.
- **Suspension:** If the "Authentication" module fails (REQ-01), all other modules are blocked and testing is suspended.

8. Suspension Criteria and Resumption Requirements

- **Suspension:** Critical "Blocker" defects (e.g., Database Connection Refused).
- **Resumption:** Successful smoke test following a new build deployment to the staging environment.

9. Test Deliverables

- 1. **RTM (Requirement Traceability Matrix):** Mapping REQ-01 through REQ-16.
- 2. **Test Case Suite:** 26 unique test cases documented in JIRA.
- 3. **Defect Log:** Detailed reports including "Steps to Reproduce," "Severity," and "Environment."
- 4. **Test Summary Report (TSR):** Final sign-off document.

10. Environmental Needs

- **Hardware:** Minimum 8GB RAM, i5 Processor.
- **Browser Stack:** Google Chrome (v120+), Mozilla Firefox (v118+).
- **Test Data:**
 - **Student Accounts:** 5 Valid, 2 Locked, 1 Unregistered.
 - **Library Data:** 100 books with varying ISBN formats and genres.

11. Responsibilities

- **Tahir Khan (QA Lead):** Oversight, RTM maintenance, and Test Plan approval.
- **QA Team:** Test Case Design (Week 4) and Execution (Week 5).
- **Dev Team:** Bug fixing and environment stability.

12. Schedule

Phase	Start Date	End Date	Resource
Requirements Analysis	April 15	April 20	Tahir Khan
Test Plan & RTM	April 21	April 29	Tahir Khan
Black-Box Design (EP/BVA)	May 01	May 07	QA Team
Execution Cycle 1	May 08	May 15	QA Team
Final Closure	May 22	May 25	Tahir Khan